

sulation. It then encapsulates the decapsulated original IP datagram in the network-level protocol data unit (PDU), for example logical link control (LLC) frame of LAN, and delivers the packet to A on FN.

Packets from MN to any other static node. Packets from MN (B) to the any other node or host (A) are sent via home agent when MN is in HN.

Packets from MN to any other node or host are sent via foreign agent when MN is in FN.

A mix of techniques described above is used from exchange of packets between two different MNs.

The principle, as stated above, will work only when MN is available to register its mobility with the foreign agent in FN. It has to register when it enters FN as well as when it leaves FN. This is done by a process called getting recognition (discovery) and registration.

Getting recognition is a process by which MN registers its location with appropriate agent, either home or foreign. MN finds a mobility agent by two methods: agent advertisement and agent solicitation message.

Agent advertisement is a process by which foreign and home agents exchange their respective presence periodically using agent advertisement message. MN in a subnet then receives agent advertisements from either its home or foreign agent. This is one way for MN to discover its location.

In second method, MN may use agent solicitation message for a quick reply for locating the agent. In either case, this process allows MN to identify its current mobility agents and then obtain one or more COAs.

Using this method, it also learns about any special services provided by a foreign agent and determines whether it is connected to its HN or FN.

Registration is a process by which having received COA, MN has to register with home agent. As MN roams, it is connected to a series of one or more FNs. The main purpose of this registration is to inform home agent of the current location of MN for correct forwarding of packets. This process provides a link between home

and COAs, and defines the lifetime of registration.

This is done by sending binding updates from MN to home agent. For this three values, namely, MN home static IP address, COA and lifetime are used. Home agent sends a registration reply to foreign agent, who forwards this reply to MN. This operation is shown in Fig. 5. Encryption is used to authenticate the registration information.

A key concern with registration is security. The default authentication algorithm uses key-based MD5 (message digest version 5) to produce a 128-bit message digest.

When MN is in FN and is registered with home agent, home agent will work as a proxy for MN. When HN receives packets addressed to MN, it intercepts those packets (using Proxy ARP) and encapsulates them with overhead COA.

Encapsulated packet is then sent to COA, which is the current IP address that foreign agent is registered with MN. Foreign agent will decapsulate those packets and forward them to MN, because it knows exactly where MN is. Encapsulation is the method used by home agent to deliver information to MN by putting an extra IP header on top of the packet.

Tunnelling is a process by which the encapsulated packet is sent to MN when it is in FN (Fig. 5). Two forms of encapsulation are specified in related standards request.

Getting recognition (discovery) makes use of ICMP. This is done by appending appropriate extension to ICMP headers. ICMP is a connectionless protocol that suits this operation. Registration requires a protocol that communicates between an application on MN and that in the home agent. Thus, it requires a transport-level protocol. As it is a simple request-and-response transaction, overhead of connection-oriented TCP is not required. Thus, UDP is used as the transport protocol for registration. Tunnelling is done at IP level. **EFY**

This article is dedicated by the author to Prof. A.K. Chowdhury, who is known as the father of computer science in India



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